The localisation of the poset model

a complement to *A model of univalence over a base category with an interval*

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Mathematical attribution. All mathematical results concerning the D modality and the D_n modalities presented here are due to Christian Sattler. This includes the constructions, proofs, and consequences. This note records their exposition and formalisation in Unit. Citations to previously published joint work retain their stated authorship.

Abstract

The poset instance of the model (`model.tex`, §8.1) is the presheaf model over finite non-empty posets. Christian Sattler's localisation uses an operation D built from the inclusion of the linear posets and the strictly monotone maps (`~/POSET/poset-model.pdf`). We present Christian Sattler's results on that construction in the setting of `model.tex`: D is a strict monad on types; if A is fibrant then so is $\mathsf{D}A$; and the two maps $\mathsf{D}A \to \mathsf{D}^2A$ are connected by a path and are homotopy equivalences. The composition of $\mathsf{D}A$ is the face-uniform filling of `model.tex`, Theorem `??`(2), first step, applied to the restriction of A to the simplex category; no excluded middle is used. The passage to a left exact modality is the criterion of Coquand, Ruch and Sattler, which is quoted, not proved.

1 Linear posets and the restriction to Δ

We work over the poset instance: $\mathcal{C}$ is the category of finite non-empty posets, $\Phi(X)$ the decidable sieves, Cl = Fib. The linear posets $[n] = \{0 < \cdots < n\}$ are objects of $\mathcal{C}$, and the simplex category Δ is the full subcategory on them; $\Delta_+ \subseteq \Delta$ has the strictly monotone maps, which are the composites of face maps. The simplicial instance of `model.tex` §8.2 is built on these same maps: its objects are the numbers n, its maps $[m] \to [n]$ the maps of $\mathcal{C}$.

Definition 1.1 (restriction)**.** For a context Γ over $\mathcal{C}$, $i^*\Gamma$ is the simplicial set $n \mapsto \Gamma([n])$; for a type A over Γ, i^*A is the type over $i^*\Gamma$ with $(i^*A)(n, \tau) = A([n], \tau)$. Restrictions are those of Γ and A along the maps of Δ.

A line of Γ at X (`model.tex` §2.2), read at the generalised points $(f, j) : [W] \to X \times [1]$ with $[W]$ linear, gives for every $f : [n] \to X$ a line of $i^*\Gamma$ at n: $(g, j) \mapsto \gamma(fg, j)$. A partial line of A over γ on a sieve φ of X gives in the same way a partial line of i^*A on the sieve $f^*\varphi$ of $[n]$. So a filling problem for A at X gives a filling problem for i^*A at every $f : [n] \to X$, and this problem is formed from the data restricted along f only.

2 Horn filling from composition

Lemma 2.1. *A fibration structure on A over Γ gives a horn filling operation on i^*A over $i^*\Gamma$ (`model.tex`, Definition of horn filling), natural in Γ.*

Proof. Let $\tau \in \Gamma([m+1])$, $0 \le k \le m+1$, and a partial element of i^*A on $\Lambda^k[m+1]$; it is given by its values a_Y on the faces Y of the horn, which are linear. If $k = 0$, let q be the constant map 0 and $g_Y : Y \times [1] \to Y$, $g_Y(y,0) = q(y)$, $g_Y(y,1) = y$. The open box of A at $[m+1]$ over the constant line at τ, with the sides $a_Y g_Y$ on the sieve generated by the faces of the horn and the lower lid $a_{\{0\}}\, q$, is composed from e_0 to e_1; the end is a filler, since $g_Y(y,1) = y$. If $k \ge 1$, the same with $q(x) = \max(x,k)$, which maps every face of the horn into itself and $[m+1]$ onto the proper face $\{x \ge k\}$, the roles of 0 and 1 exchanged, and the composition from e_1 to e_0. The construction uses only the data and the composition, so it is natural. □

This is Theorem **??**(1) of `model.tex` with the retraction made explicit, here for a type over posets read on the linear posets.

3 The operation D

Definition 3.1. Let A be a type over Γ. For $\rho \in \Gamma(X)$, $\mathsf{D}A(X,\rho)$ is the set of families $u(f) \in A([n], \rho f)$, for all n and $f : [n] \to X$, with

$$u(f)\, g = u(fg) \qquad \text{for every strictly monotone } g : [m] \to [n].$$

Restriction along $k : X' \to X$ is $(uk)(f) = u(kf)$. For $\alpha : A \to B$, $(\mathsf{D}\alpha\, u)(f) = \alpha(u(f))$.

It suffices to require the condition for the face maps. The family and the condition are coded (a family indexed by n and the maps $[n] \to X$, a proposition), so $\mathsf{D}A$ is a type. D commutes with substitution: for $s : \Delta \to \Gamma$, $\mathsf{D}(A[s])$ maps to $\mathsf{D}A$ by a cartesian map over s whose fibre maps transport the values along the naturality of s, as for Π (`model.tex`, Theorem **??**).

Definition 3.2. $\eta_A : A \to \mathsf{D}A$, $(\eta_A a)(f) = af$; and $\mu_A : \mathsf{D}^2 A \to \mathsf{D}A$, $(\mu_A v)(f) = v(f)(\mathrm{id})$.

Lemma 3.3 (strict monad)**.** *η and μ are natural, $\mathsf{D}\alpha \circ \mu_A = \mu_B \circ \mathsf{D}^2\alpha$, and*

$$\mu_A \circ \eta_{\mathsf{D}A} = \mathrm{id}, \qquad \mu_A \circ \mathsf{D}\eta_A = \mathrm{id}, \qquad \mu_A \circ \mathsf{D}\mu_A = \mu_A \circ \mu_{\mathsf{D}A},$$

as equalities of maps, that is of codes.

Proof. Unfold the definitions (poset-model, Lemma 7.4). □

4 The composition of DA

Theorem 4.1. *A horn filling operation on i^*A gives a fibration structure on $\mathsf{D}A$. Taken on the universal family, it is natural in Γ. With Lemma 2.1: if A is fibrant, so is $\mathsf{D}A$.*

Proof. Let γ be a line of Γ at X, $u_0 \in \mathsf{D}A(\gamma(e))$, and (φ, u) a partial line of $\mathsf{D}A$ ending at u_0. For $f : [n] \to X$ the data read along f (Section 1) form a filling problem for i^*A at n: the line $(g,j) \mapsto \gamma(fg, j)$, the end $u_0(f)$, and the partial line on $f^*\varphi$ whose value at (g,j) is u_{fg} at the corresponding chain. These values are natural only along the injective maps of Δ; they are extended to a simplicial partial line by the Eilenberg–Zilber factorisation: factor $\langle g, j\rangle = m\sigma$ with m an injective chain of the prism and σ surjective, and take the value at m restricted along σ. The factorisation is that of the composite point of the prism, not of a stage of the partial line: a degenerate restriction of a stage would otherwise be read through a non-injective map. This is natural along every map, since the factorisation is unique and $v(m)m' = v(mm')$ for m' injective,

and it commutes with restriction along face maps; only its values on chains are used below. Let $F(f)$ be its *face-uniform* filler (`model.tex`, proof of Theorem **??**(2), first step: fill the shuffles $P_n, \dots, P_0$ one after the other, each by one horn filling, and take the side when it contains the identity). The composite is $f \mapsto F(f)(e')$.

- It is in $\mathsf{D}A$: a face of the filler at f is the filler at the face of f, since the face-uniform fillers are uniform along face maps, and faces suffice.
- It agrees with the sides: if f lands in φ, the side of the problem contains the identity and the filler is that side.
- It is uniform along every $k : X' \to X$: the problem of the restricted data at f is the problem of the given data at kf, so the two fillers are the same.

The horn filling operation is only used on the prisms $[n] \times \Delta[1]$, so it needs no uniformity, and the face-uniform fillers need no excluded middle.

Naturality is obtained as for Ext (`model.tex`, §6.2), by making the construction once on the universal family. El is fibrant (`model.tex`, §8.1), so DEl over U has the fibration just built. Every fibrant type of the poset model is $c^*\mathsf{El}$ for its code map $c : \Gamma \to \mathsf{U}$; its D carries the fibration of DEl moved along the cartesian map $\mathsf{D}(c^*\mathsf{El}) \to \mathsf{DEl}$ over c (Section 3). For $s : \Delta \to \Gamma$ the cartesian maps compose, so the structure on $\mathsf{D}(A[s])$ is the move of that on $\mathsf{D}A$ by the composition law of moves (`model.tex`, Lemma **??**); the fillers themselves are never moved. □

Remark 4.2. Only the uniformity along the face maps is used. This is why $\mathsf{D}A$ is fibrant without the normalised fillers, which require excluded middle and make fillers uniform along the degeneracies as well: the condition defining $\mathsf{D}A$ only concerns Δ_+.

5 Paths and the two maps $\mathsf{D}A \to \mathsf{D}^2A$

A *homotopy* between maps $\alpha, \beta : A \to B$ over Γ is an element of the path type $\mathrm{Path}_B(\alpha, \beta)$ over $\Gamma.A$ (`model.tex`, Theorem **??**): at (ρ, a), a line in B over the constant line at ρ from $\alpha(a)$ to $\beta(a)$, natural in (ρ, a). We write $\alpha \simeq \beta$; a *homotopy equivalence* is a map with a two-sided homotopy inverse. Homotopies compose with maps on both sides.

Lemma 5.1. *If $\alpha \simeq \beta$ then $\mathsf{D}\alpha \simeq \mathsf{D}\beta$.*

Proof. For a generalised point $F = (f, h) : [n] \to X \times [1]$ put $H'(u)(F) = H(u(f))\langle \mathrm{id}, h\rangle$, where $\langle \mathrm{id}, h\rangle : [n] \to [n] \times [1]$ (poset-model, Lemma 9.2). □

Lemma 5.2. *$\eta_{\mathsf{D}A} \simeq \mathsf{D}\eta_A$, by a homotopy which uses no composition or filling of A.*

Proof. For $F = (f, h)$, h cuts $[n] = A_0 + A_1$; a map $g : [m] \to [n]$ cuts $[m] = B_0 + B_1$, with $g = (g_0 + A_1)(B_0 + g_1)$. Put $v(F)(g) = u\big(f(g_0 + A_1)\big)(B_0 + g_1)$. Every condition on v is an instance of one principle: $u(F)K$ is unchanged when (F, K) is reparametrised along a strictly monotone map (poset-model, Lemma 9.3). □

Corollary 5.3. *$\mu_A \circ \mathsf{D}\eta_A = \mathrm{id}$ and $\mathrm{id} \simeq \mathsf{D}\eta_A \circ \mu_A$, with no fibrancy assumption. If A is fibrant, $\mathsf{D}\eta_A$ and $\eta_{\mathsf{D}A}$ are homotopy equivalences.*

Proof. $\mu_A \circ \mathsf{D}\eta_A = \mathrm{id}$ by Lemma 3.3. By Lemma 5.2 and Lemma 5.1, $\mathsf{D}\eta_{\mathsf{D}A} \simeq \mathsf{D}^2\eta_A$; composing with $\mu_{\mathsf{D}A}$ on the left and using Lemma 3.3, $\mathrm{id} = \mu_{\mathsf{D}A} \circ \mathsf{D}\eta_{\mathsf{D}A} \simeq \mu_{\mathsf{D}A} \circ \mathsf{D}^2\eta_A = \mathsf{D}\eta_A \circ \mu_A$. For $\eta_{\mathsf{D}A}$: $\mu_A \circ \eta_{\mathsf{D}A} = \mathrm{id}$, and $\eta_{\mathsf{D}A} \circ \mu_A \simeq \mathsf{D}\eta_A \circ \mu_A$ by Lemma 5.2. When A is fibrant so is D^2A (Theorem 4.1 twice), and homotopies into a fibrant type can be inverted and composed, by composition in the target; so $\eta_{\mathsf{D}A} \circ \mu_A \simeq \mathrm{id}$ and $\mathsf{D}\eta_A \circ \mu_A \simeq \mathrm{id}$. □

6 The universal property of η

A fibrant type B is D-*modal* when η_B is a homotopy equivalence. By Corollary 5.3, $\mathsf{D}A$ is D-modal for every fibrant A (condition 1 of the criterion below).

Proposition 6.1. *Let A be a type and B a fibrant D-modal type over Γ, with $p_B : \mathsf{D}B \to B$ and $p_B \circ \eta_B \simeq \mathrm{id}$. For $u : A \to B$ put $G(u) = p_B \circ \mathsf{D}u : \mathsf{D}A \to B$. Then $G(u) \circ \eta_A \simeq u$, and $G(f \circ \eta_A) \simeq f$ for every $f : \mathsf{D}A \to B$. So precomposition with η_A is a bijection up to homotopy between the maps $\mathsf{D}A \to B$ and the maps $A \to B$.*

Proof. $G(u) \circ \eta_A = p_B \circ \mathsf{D}u \circ \eta_A = p_B \circ \eta_B \circ u \simeq u$, by the naturality of η (strict). And $G(f \circ \eta_A) = p_B \circ \mathsf{D}f \circ \mathsf{D}\eta_A \simeq p_B \circ \mathsf{D}f \circ \eta_{\mathsf{D}A} = p_B \circ \eta_B \circ f \simeq f$, by Lemma 5.2 and the naturality of η. The homotopies are composed in the fibrant type B. □

This is the argument of [CRS], Theorem 1.10, (2)⇒(1), for maps over Γ rather than for the internal function types.

7 What is quoted, and what may follow

With Theorem 4.1, Lemma 5.2, Corollary 5.3 and Lemma 3.3, the criterion of Coquand, Ruch and Sattler (poset-model, Theorem 10.1) makes D a left exact modality, and the D-modal types form a constructive model of univalent type theory with higher inductive types. That criterion is not formalised. Two further results of poset-model use only the structures above and could be added: a proposition with a point over [0] has an element of $\mathsf{D}A$ (Proposition 11.4), and $\mathsf{D}A$ is trivially fibrant when spheres bound in A (Theorem 12.3, the trivial case of Whitehead's principle).

A Correspondence with the formalisation

Layer L9 of `seg/core`, over the poset instance; all statements of this note are formalised except those of the last section.

statement	files
Section 1	CLocRestr
Lemma 2.1	CLocHorn
Definitions of D, η, μ, Lemma 3.3	CLocD
Theorem 4.1	CLocEZ (the factorisation), CLocCompV, CLocCompP, CLocComp; naturality on the universal family: CLocDCart, CLocDUniv
homotopies, Lemma 5.1	CHtpyOps (L3: homotopies, whiskering, inverse and composition into a fibrant type), CLocHtpy
Lemma 5.2	CLocPath, CLocPathEnd, CLocEtaH
Corollary 5.3	CLocEq
Proposition 6.1	CLocUniv (isModal, dModal; UP.PA.resA, UP.PBm.resB)